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INFORMATION
TECHNOLOGY



FIT3179 Data Visualisation

Week 10: Animation – Discussion

Homework Week 10

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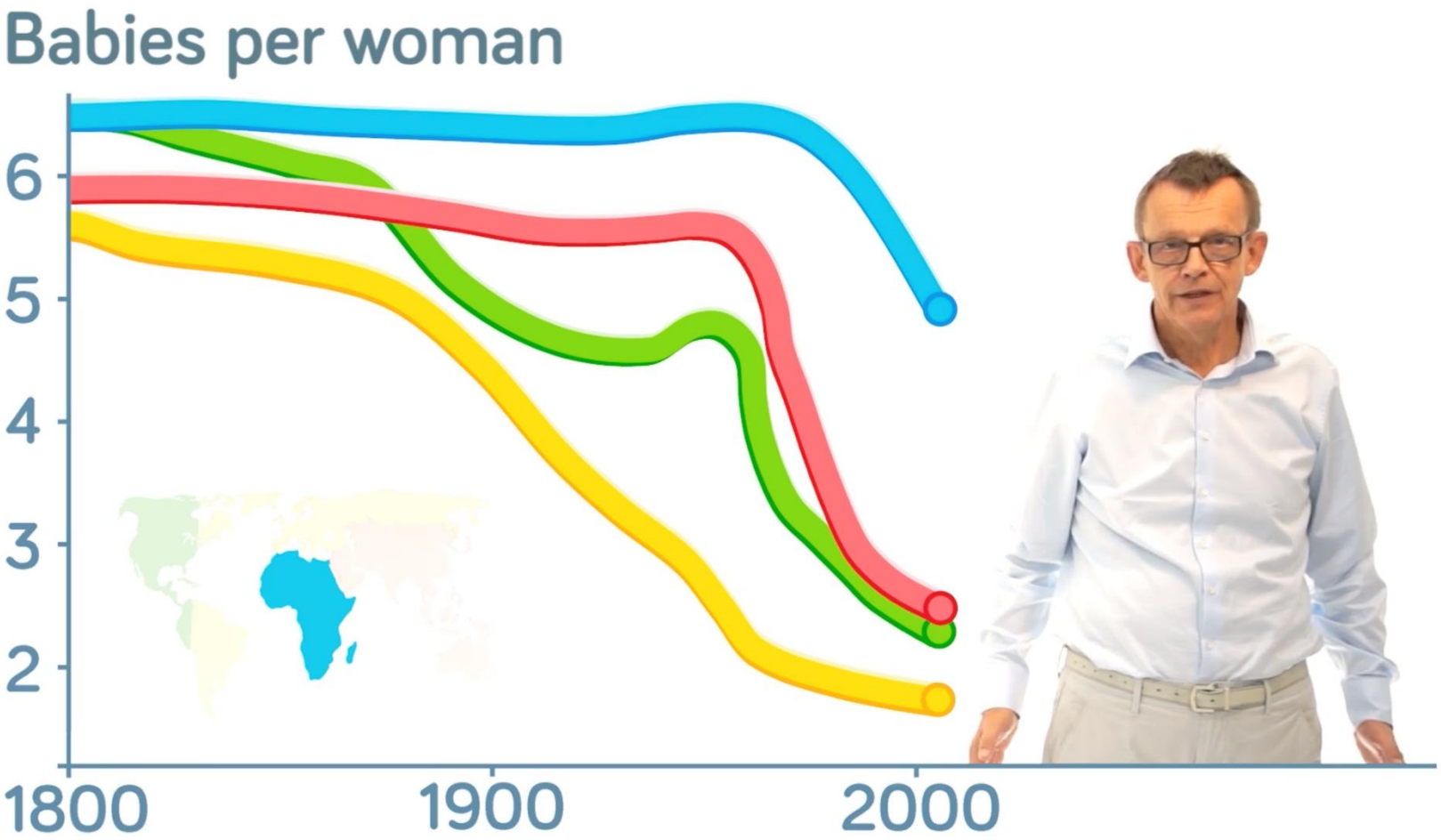
- DV1 marking, to be released
- Data Visualisation 2 (DV2)
- Studios.
  - Vega-Lite advance interactivity
  - Test 2. week 11.
    - Sample questions will be released in week 10.

Questions?

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- Any questions about the pre-recorded video lectures?

1. Animation
    1. Frames rendered by the browser or encoded in video
    2. Temporal animation: change marks and channels in time
    3. Non-temporal animation: animate between idioms
    4. Cognitive load, short, and long-term memory
    5. Change blindness
    6. Directing user attention with annotation and modifying saliency
2. Visualisation Tools



How Did Babies per Woman Change in Different Regions? by Hans Rosling for gapminder.org

Questions to ask when evaluating animation for data visualisation

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What is animated?

- **How many items** are animated?
- What are the **animated channels and marks**?
- Are **data visualisation principles** applied correctly?
- What **type of change** is the animation showing? For example, movement, processes or algorithms, storytelling, time, zoom for showing relative size.

Short-term memory and change blindness

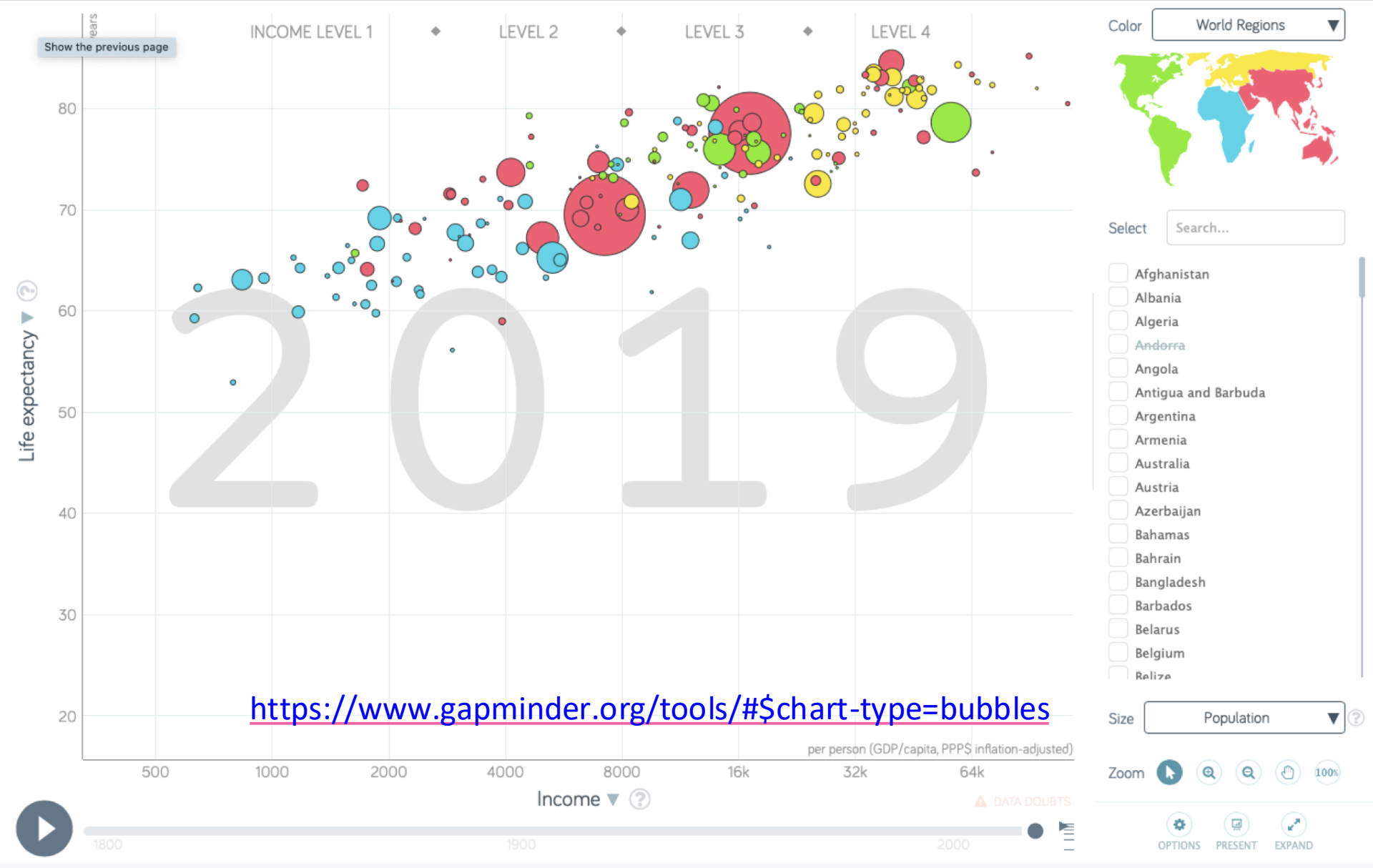
- What techniques are used to **reduce cognitive load**?
- What aspects rely on **short-term working memory**?
- Is **change blindness** an issue?
- Is the visualisation **guiding user attention** and focus?

Speculative assessment

- Do you think…
  - the **main message is made permanent in long term memory**?
  - the **cognitive load** is too high / high / about right / low?
  - the use of animation is **overall good or bad**?

Temporal Animation

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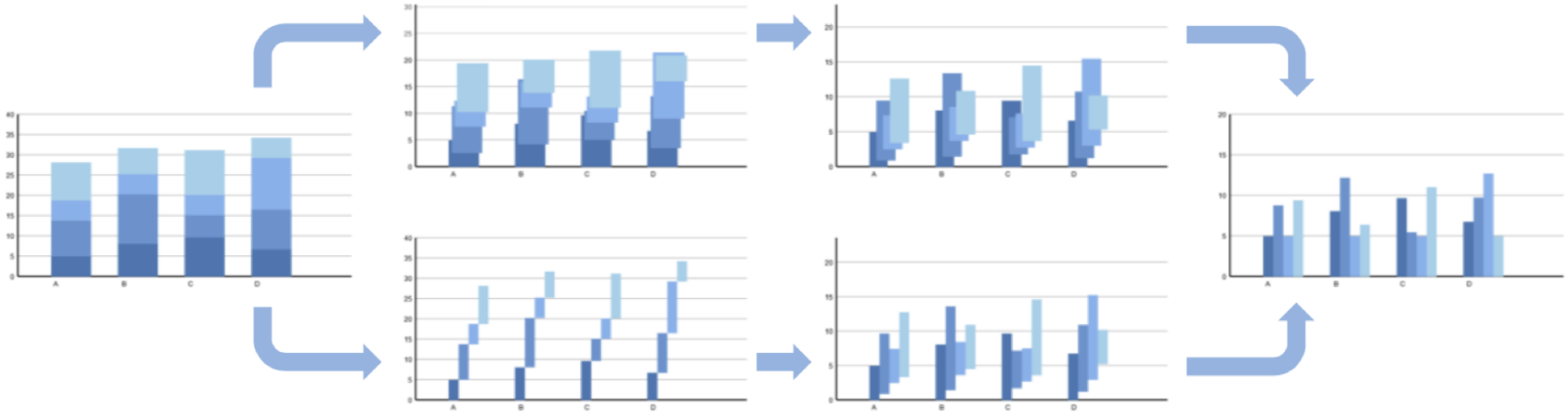
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Non-temporal Animation



Heer and Robertson 2007
Animated Transitions in Statistical Data Graphics

https://www.youtube.com/watch?v=vLk7mlAtEXI

When to use animation?

It would be a poor idea to use animation for complex data analysis or a lookup/locate action.



Animation may be useful for data presentation.

Change Blindness

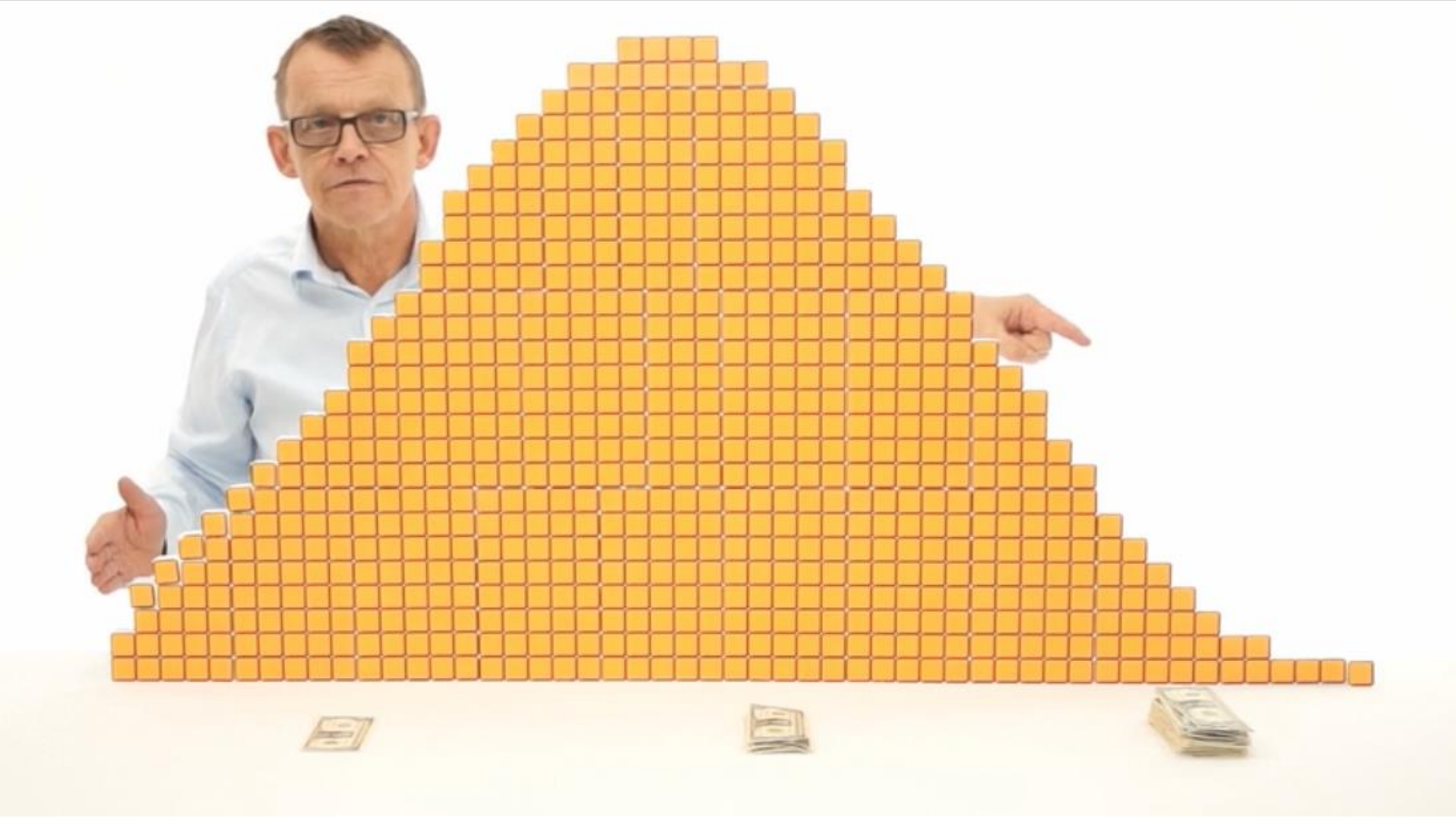
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https://www.youtube.com/watch?v=IGQmdoK_ZfY

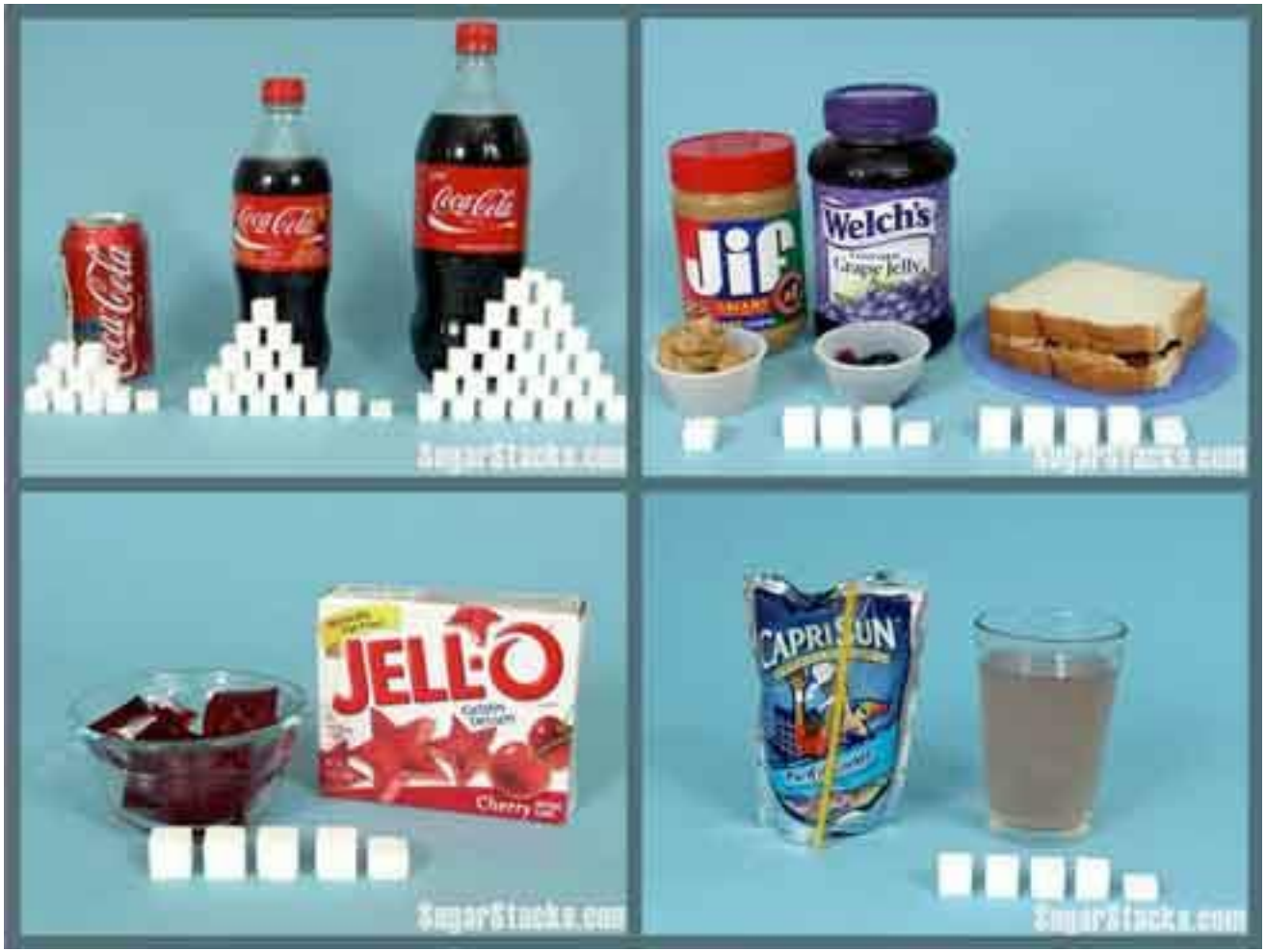
Data Story Telling with Physical Props

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https://www.gapminder.org/answers/how-many-are-rich-and-how-many-are-poor/

Data Physicalisation



Data Physicalisation

dataphys.org/list/gallery/

Gallery of Physical Visualizations

and Related Artifacts

This is a chronological list of physical visualizations and related artifacts, maintained by Pierre Dragicevic and Yvonne Jansen. Thanks to our contributors. Feel free to post a general comment or if you know of another interesting physical visualization, please submit it!

This list currently has 351 entries. See recent additions. You can also get notified of new entries through Twitter.

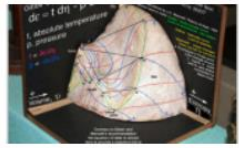
List view | Passive physical visualizations (204) | Active physical visualizations (39) | Physical models (34) | Measuring instruments (11) | Interactive installations (7) | Enabling technologies (26) | Other (7) | Uncertain (23)


5500 BC – Mesopotamian Clay Tokens


2600 BC – Inca Quipus

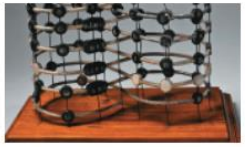

500 BC – Visualizing Opinions with Pebbles


1862 – Marshall Islands Stick Charts

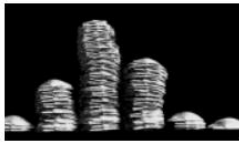

1871 – Thermodynamic Surfaces


1890 – Polynesian Genealogical Instrument

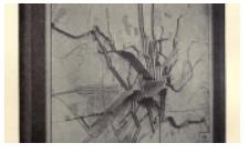

1898 – Tallies Used as Social Displays on Pacific...

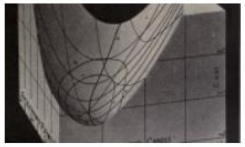

1898 – Crookes' Vis Generatrix

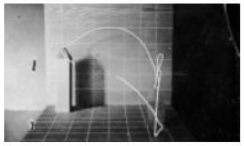

1900 – Pearson and Lee's Height Correlation...


1901 – Davenport's Physical Distributions

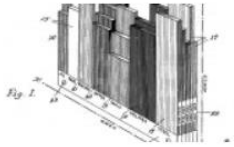

1907 – Pin Maps


1913 – Frankfurt Streetcar Load

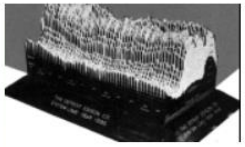

1914 – Solid 3D Curves for Engineering


1915 – Wire Models of Factory Worker Movements

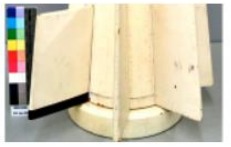

1920 – Yakama Time Ball


1926 – Karsten's Tridimensional Chart


1933 – IBM's Cosmograph


1935 – 3D Visualizations of Power Consumption


1939 – Map of Great Britain's Marine Trade


1940s – Stedman's 3D Periodic Table